EE6320 RF Integrated Circuits
Project: PA Design

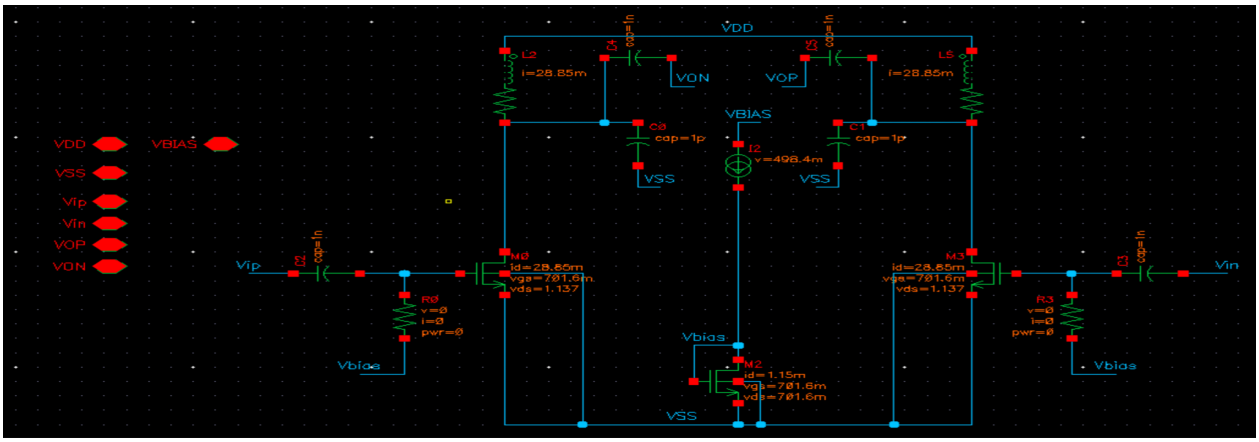
PA Performance Summary Table

Design Parameter	Design Metric	Performance	Specification
Output P1dB	f_o = 3450 MHz	13.06 dBm	$\geqslant$ +13 dBm
	f_o = 3500 MHz	13.058 dBm	$\geqslant$ +13 dBm
	f_o = 3550 MHz	13.0472 dBm	$\geqslant$ +13 dBm
AM-PM Deviation (at P1dB)	f_o = 3450 MHz	0.5 degrees	$\leqslant$ 3 degrees
	f_o = 3500 MHz	0.63 degrees	$\leqslant$ 3 degrees
	f_o = 3550 MHz	0.85 degrees	$\leqslant$ 3 degrees
Voltage Gain (from Gate to Drain)	f_o = 3450 MHz	4.354	$\geqslant$ 2
	f_o = 3500 MHz	4.355	$\geqslant$ 2
	f_o = 3550 MHz	4.357	$\geqslant$ 2
Power (at 3.5 GHz)	PA Average Consumption (excluding bias)	69.25 mW	Minimize
	Bias Circuit Consumption	1.38 mW	Minimize
Other	Sum of all Capacitances (including ac coupling)	4*1n + 2*1p = 4.002n	-

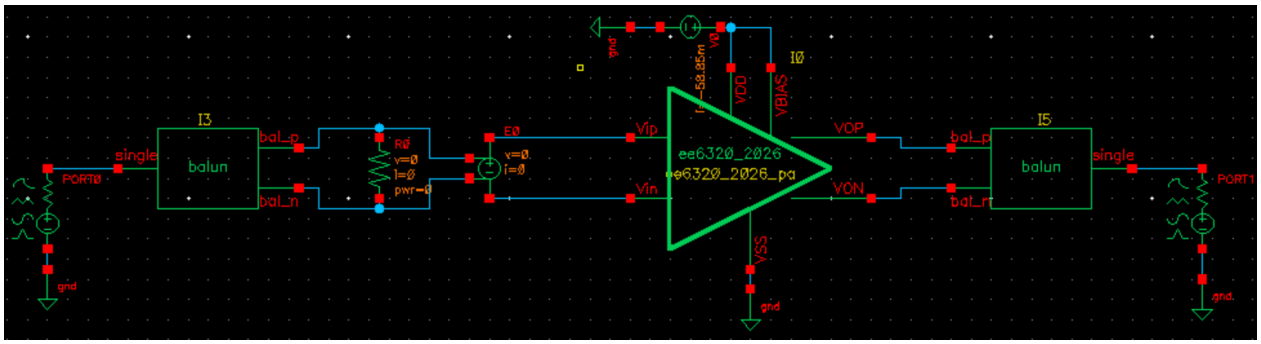
	Inductance Used	**2n*2 = 4n**	**-**
	Simulator used	**ADE L**	

Name:N Yasho Srikar Adithya Reddy
Roll No: EE23B114

PA Schematic



PA Testbench



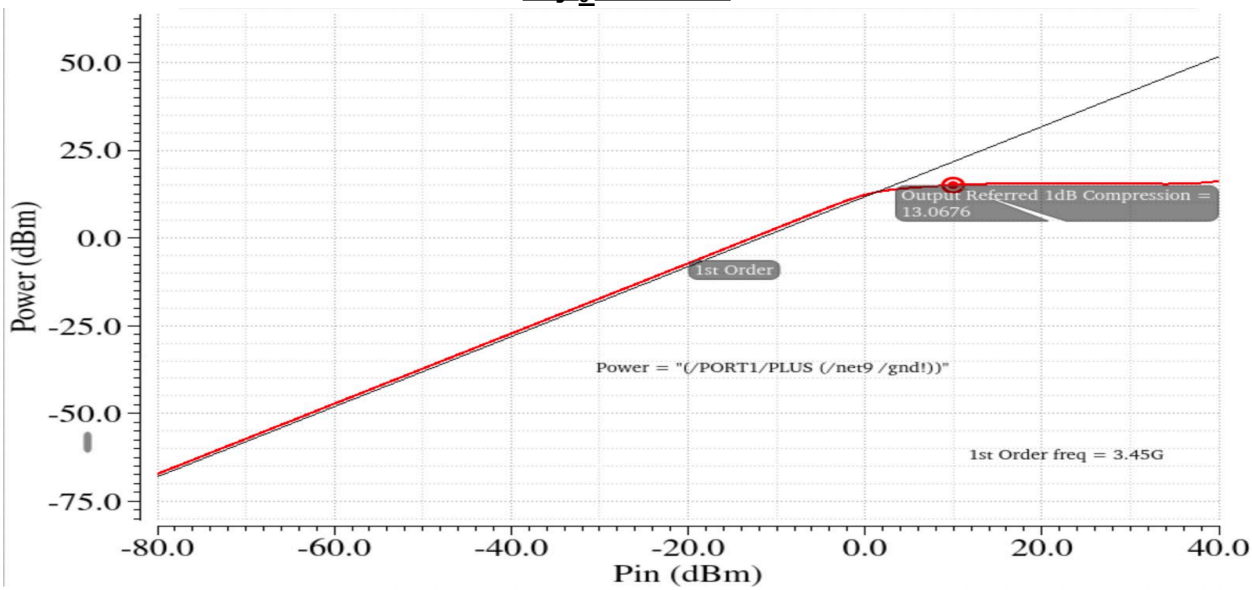
Component Values Table

(W, L) MOS Amplifier = 200u, 120n	**(L, C) Drain Tank = (2n*2, 1p*2)**
(W, L) Current Mirror MOS = 10u, 120n	**Bias Current = 1.15 mA**

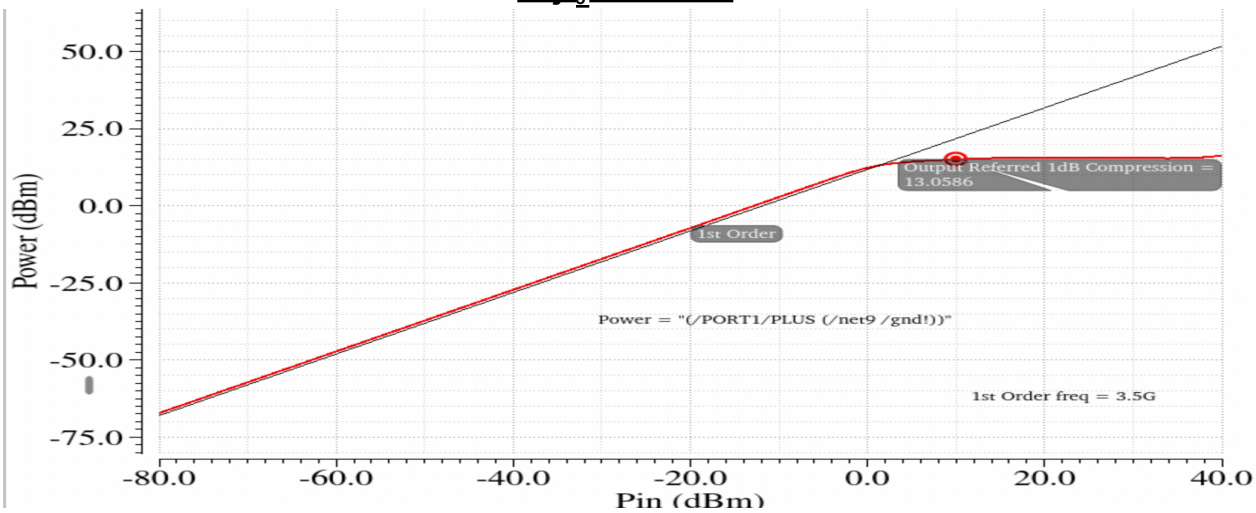
R_bias = 10 KΩ, C_coupling =1 nF	V_DD = 1.2 , R_load = 50 ohm

Output 1dB Compression Point Plots

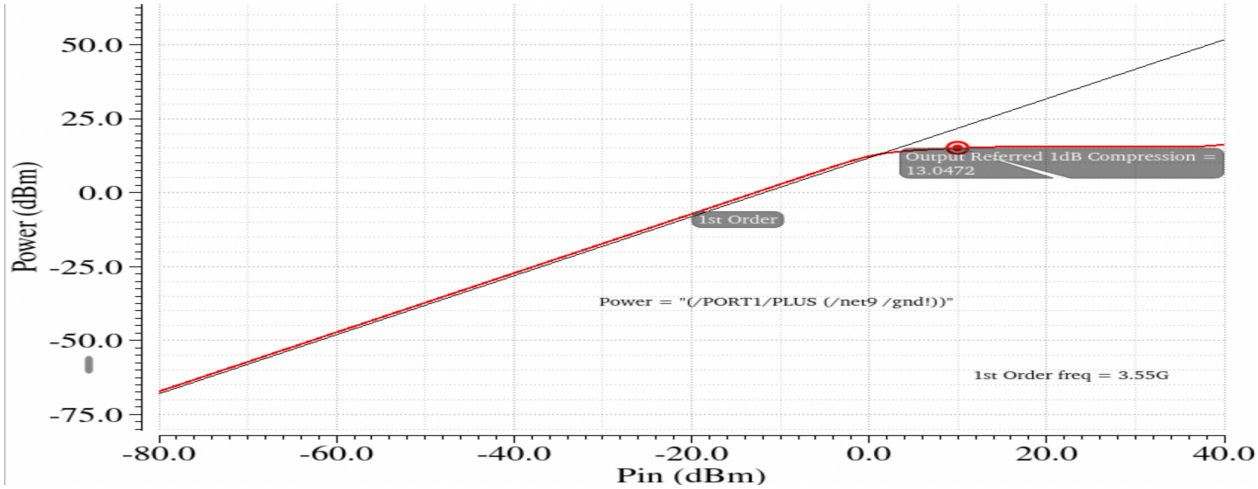
At f_o =3450 MHz



At f_o = 3500 MHz

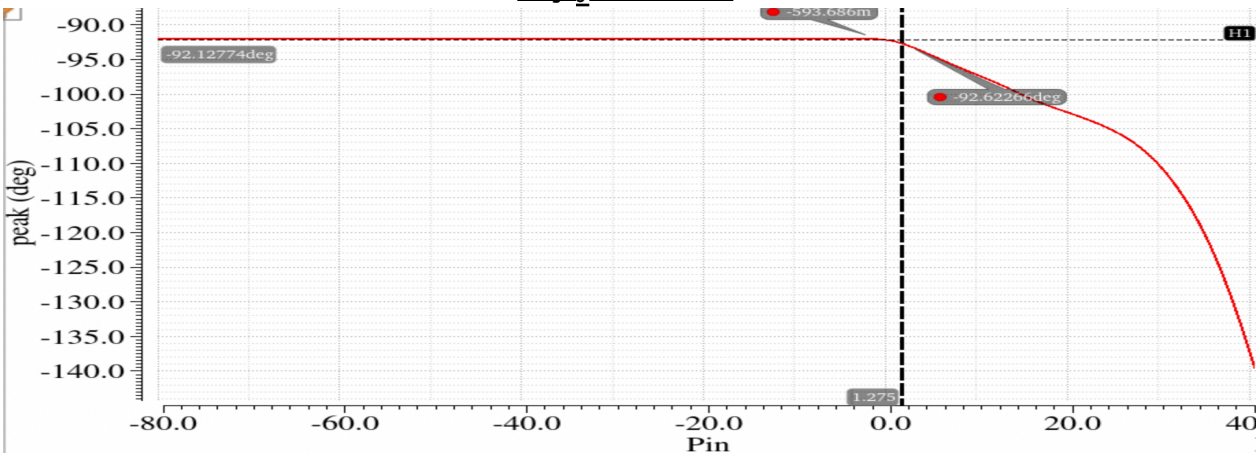


At f_o = 3550 MHz

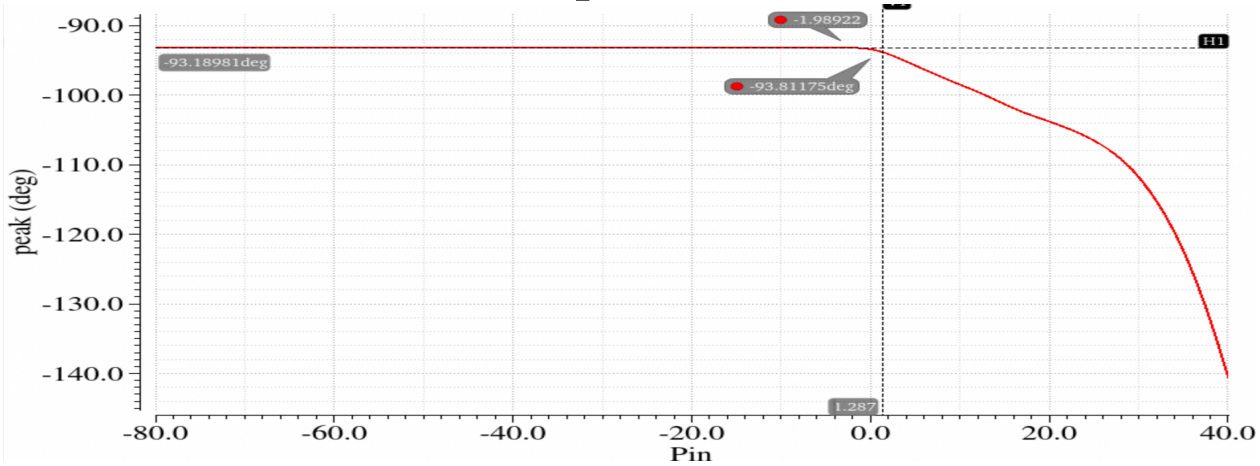


AM-PM Deviation Plots

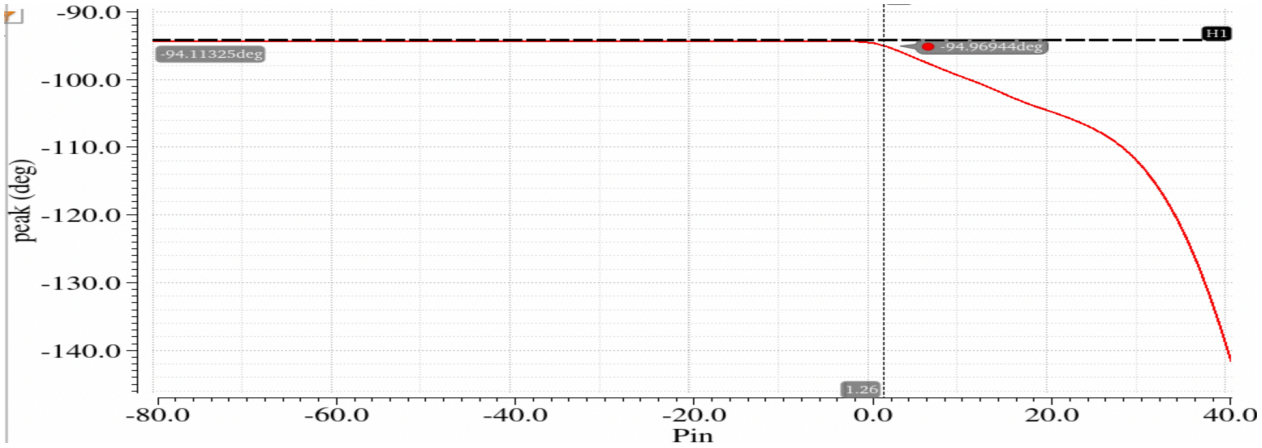
At f_o = 3450 MHz



At f_o = 3500 MHz

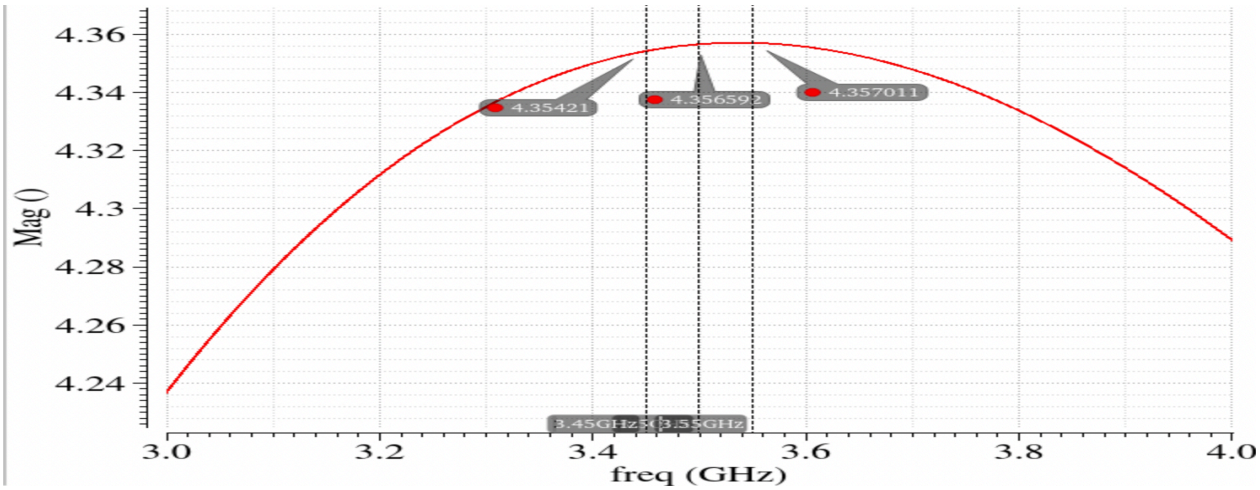


At f_o =3550 MHz



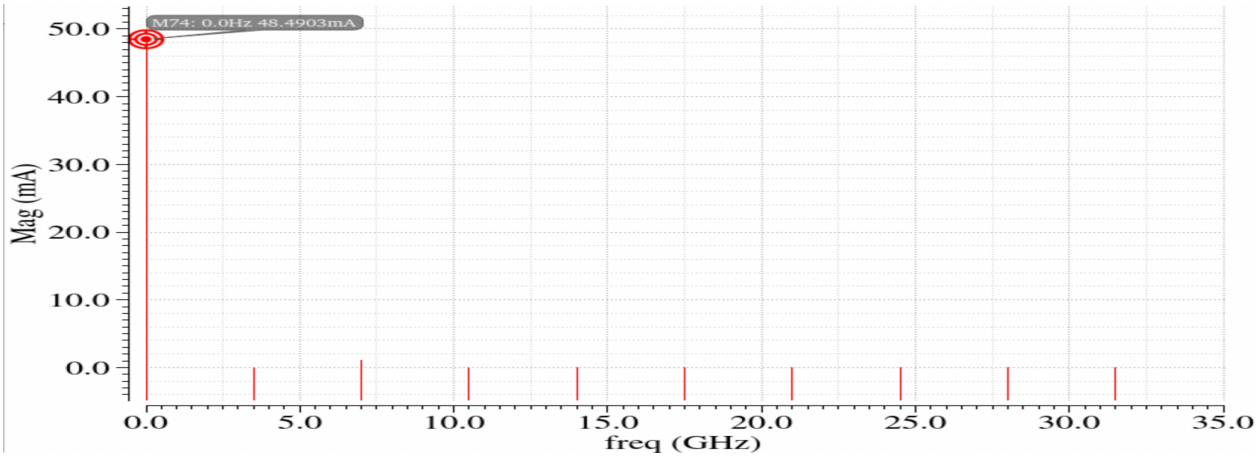
PA Voltage Gain Plot

(from Gate to Drain)



Current through VDD Plot

(f_o = 3500 MHz)



Design Procedure

Started with MOS characterisation for 1u width mos to find gm3 zero point which I found to be 200uA/um current density. Now I used that and started to increase multipliers until I reached the required specs. After that, I tuned the capacitor for the proper gain band and tuned the bias circuit.

Path to Project Files

/home/ee23b114/cadence/EE6320_2026/ee6320_2026/PA_TB
/home/ee23b114/cadence/EE6320_2026/ee6320_2026/ee6320_2026_pa

Note: All the schematic/plots put up here are for reference only. Please replace them with the respective plots/schematic used or obtained by you.